

Table 1: Pass-sum execution time (s, median per iteration; sum of pass medians, not full executable wall time) and speedup split by pass type. When present, the OctTree row includes ColorOctree passes; a separate ColorOctree row reports only those passes. ADT fields = non-recursive fields annotated with `@BENCH adt_fields`; SoA bufs = 1+ non-recursive field slots across all constructors (recursive children are stored in the tag buffer). Speedup $>1\times$ means the denominator is faster; **bold** marks $>1.1\times$.

Program	ADT fields	SoA bufs	End-to-end					Fold passes					Map passes				
			Am (s)	Ai (s)	Sm (s)	Am/Sm	Ai/Sm	Am (s)	Ai (s)	Sm (s)	Am/Sm	Ai/Sm	Am (s)	Ai (s)	Sm (s)	Am/Sm	Ai/Sm
OctTree	16	9	0.6880	–	0.6781	$1.01\times$	–	0.3851	–	0.3282	$1.17\times$	–	0.3029	–	0.3499	$0.87\times$	–
Compiler	9	8	0.2845	1.091	0.2191	$1.30\times$	$4.98\times$	0.1264	0.7190	0.0394	$3.21\times$	$18.24\times$	0.1582	0.3715	0.1796	$0.88\times$	$2.07\times$
DBQuery	15	13	0.2564	0.3244	0.2526	$1.02\times$	$1.28\times$	0.1188	0.1736	0.0870	$1.37\times$	$2.00\times$	0.1376	0.1508	0.1656	$0.83\times$	$0.91\times$
DecisionTree	7	6	3.695	5.517	2.950	$1.25\times$	$1.87\times$	3.695	5.517	2.950	$1.25\times$	$1.87\times$	–	–	–	–	–
DomTree	15	14	0.1956	0.3351	0.1426	$1.37\times$	$2.35\times$	0.1378	0.2751	0.0640	$2.15\times$	$4.30\times$	0.0578	0.0600	0.0786	$0.74\times$	$0.76\times$
KDTree	17	16	0.1796	0.2460	0.1394	$1.29\times$	$1.77\times$	0.1796	0.2460	0.1394	$1.29\times$	$1.77\times$	–	–	–	–	–
LinearListReduction	11	11	0.0297	0.1349	0.0050	$5.95\times$	$27.10\times$	0.0297	0.1349	0.0050	$5.95\times$	$27.10\times$	–	–	–	–	–
List	2	2	0.5729	–	0.6407	$0.89\times$	–	0.3231	–	0.3020	$1.07\times$	–	0.2498	–	0.3387	$0.74\times$	–
MonoTree	3	2	0.0635	0.1407	0.0652	$0.97\times$	$2.16\times$	0.0177	0.0805	0.0227	$0.78\times$	$3.54\times$	0.0459	0.0603	0.0425	$1.08\times$	$1.42\times$
ObjectGraph	5	4	0.2394	0.4814	0.2419	$0.99\times$	$1.99\times$	0.0973	0.3064	0.0806	$1.21\times$	$3.80\times$	0.1421	0.1751	0.1613	$0.88\times$	$1.09\times$
PiecewiseFunctions	8	7	0.3855	0.5640	0.3472	$1.11\times$	$1.62\times$	0.1487	0.3022	0.0901	$1.65\times$	$3.35\times$	0.2368	0.2618	0.2572	$0.92\times$	$1.02\times$
TernaryTree	5	3	0.1134	0.1436	0.1062	$1.07\times$	$1.35\times$	0.0299	0.0506	0.0288	$1.04\times$	$1.76\times$	0.0835	0.0930	0.0774	$1.08\times$	$1.20\times$
Trie	10	9	0.2172	0.2974	0.2018	$1.08\times$	$1.47\times$	0.0678	0.1373	0.0321	$2.12\times$	$4.28\times$	0.1493	0.1601	0.1698	$0.88\times$	$0.94\times$
ColorOctree	25	18	0.0960	0.1219	0.0768	$1.25\times$	$1.59\times$	0.0960	0.1219	0.0768	$1.25\times$	$1.59\times$	–	–	–	–	–

Table 2: Baseline Gibbon comparison (times only). Shown variants: AoS-mut, AoS-imm, SoA-mut. Times are full executable end-to-end wall time per run (s), not pass-sum time. **Bold** marks the fastest time across shown variants.

Program	Am	Ai	Sm
Compiler	30.20	110.4	23.37
DBQuery	26.71	33.74	26.51
DecisionTree	369.0	551.5	296.7
DomTree	20.39	34.22	15.06
KDTree	18.14	24.79	14.33
LinearListReduction	3.252	14.34	0.8513
List	64.35	<i>OOM</i>	68.08
MonoTree	6.833	14.90	6.881
ObjectGraph	24.90	49.07	25.37
OctTree	64.49	69.85	69.70
PiecewiseFunctions	40.46	58.45	36.85
TernaryTree	11.87	14.99	11.47
Trie	22.98	31.03	21.49
ColorOctree	9.894	12.74	8.490

Table 3: Baseline Gibbon comparison (speedups). Shown: AoS-mut/SoA-mut, AoS-imm/AoS-mut, AoS-imm/SoA-mut. Speedups are computed from full executable end-to-end wall time per run. $>1\times$ means the denominator is faster.

Program	Am/Sm	Ai/Am	Ai/Sm
Compiler	$1.29\times$	$3.66\times$	$4.72\times$
DBQuery	$1.01\times$	$1.26\times$	$1.27\times$
DecisionTree	$1.24\times$	$1.49\times$	$1.86\times$
DomTree	$1.35\times$	$1.68\times$	$2.27\times$
KDTree	$1.27\times$	$1.37\times$	$1.73\times$
LinearListReduction	$3.82\times$	$4.41\times$	$16.85\times$
List	$0.95\times$	–	–
MonoTree	$0.99\times$	$2.18\times$	$2.17\times$
ObjectGraph	$0.98\times$	$1.97\times$	$1.93\times$
OctTree	$0.93\times$	$1.08\times$	$1.00\times$
PiecewiseFunctions	$1.10\times$	$1.44\times$	$1.59\times$
TernaryTree	$1.04\times$	$1.26\times$	$1.31\times$
Trie	$1.07\times$	$1.35\times$	$1.44\times$
ColorOctree	$1.17\times$	$1.29\times$	$1.50\times$

Table 4: Per-pass performance for `OctTree`, OctTree SoA uses 9 buffers; ColorOctree SoA uses 18 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across `-iterate` runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Am/Sm	Ai/Am	Ai/Sm
barnesHutPotential	F	11/16	31%	0.0439 $\pm$ 0.00107	0.0591 $\pm$ 0.00001	0.0394$\pm$0.00095	1.11$\times$	1.35$\times$	1.50$\times$
countActive	F	10/16	38%	0.0362 $\pm$ 0.00001	0.0466 $\pm$ 0.00002	0.0291$\pm$0.00117	1.24$\times$	1.29$\times$	1.60$\times$
countParticles	F	8/16	50%	0.0368 $\pm$ 0.00001	0.0448 $\pm$ 0.00001	0.0265$\pm$0.00001	1.39$\times$	1.22$\times$	1.69$\times$
fmmPotential	F	12/16	25%	0.0980 $\pm$ 0.00114	0.0982 $\pm$ 0.00002	0.0958$\pm$0.00116	1.02 $\times$	1.00 $\times$	1.02 $\times$
sumEnergy	F	12/16	25%	0.0384 $\pm$ 0.00001	0.0547 $\pm$ 0.00123	0.0333$\pm$0.00003	1.15$\times$	1.42$\times$	1.64$\times$
sumMass	F	10/16	38%	0.0357 $\pm$ 0.00154	0.0442 $\pm$ 0.00002	0.0272$\pm$0.00001	1.31$\times$	1.24$\times$	1.62$\times$
clearFlags	M	15/16	6%	0.1528 $\pm$ 0.00013	0.1495$\pm$0.00024	0.1744 $\pm$ 0.00116	0.88 $\times$	0.98 $\times$	0.86 $\times$
scaleEnergy	M	16/16	0%	0.1501$\pm$0.00189	0.1513 $\pm$ 0.00162	0.1755 $\pm$ 0.00016	0.86 $\times$	1.01 $\times$	0.86 $\times$
paletteEntriesQuantized	F	13/25	48%	0.0465 $\pm$ 0.00002	0.0556 $\pm$ 0.00002	0.0336$\pm$0.00004	1.38$\times$	1.20$\times$	1.65$\times$
quantizationErrorProxy	F	10/25	60%	0.0496 $\pm$ 0.00016	0.0663 $\pm$ 0.00172	0.0432$\pm$0.00121	1.15$\times$	1.34$\times$	1.54$\times$
Total				0.6880	0.7702	0.6781	1.01 $\times$	1.12$\times$	1.14$\times$
Geomean						1.13$\times$	1.19$\times$	1.35$\times$	

Table 5: Per-pass performance for `Compiler`, ADT has 9 fields, SoA uses 8 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across `-iterate` runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Am/Sm	Ai/Am	Ai/Sm
blockCountPass	F	2/9	78%	0.0120 $\pm$ 0.00043	0.0788 $\pm$ 0.00003	0.0019$\pm$0.00000	6.24$\times$	6.58$\times$	41.08$\times$
branchStatsPass	F	2/9	78%	0.0117 $\pm$ 0.00001	0.0788 $\pm$ 0.00002	0.0034$\pm$0.00000	3.46$\times$	6.71$\times$	23.18$\times$
castInstCountPass	F	3/9	67%	0.0181 $\pm$ 0.00001	0.0788 $\pm$ 0.00132	0.0013$\pm$0.00000	13.96$\times$	4.36$\times$	60.92$\times$
has cycle	F	4/9	56%	0.0244 $\pm$ 0.00113	0.0923 $\pm$ 0.00114	0.0177$\pm$0.00015	1.38$\times$	3.78$\times$	5.20$\times$
instCountPass	F	2/9	78%	0.0119 $\pm$ 0.00001	0.0787 $\pm$ 0.00114	0.0019$\pm$0.00000	6.17$\times$	6.62$\times$	40.87$\times$
latencyModelPass	F	3/9	67%	0.0120 $\pm$ 0.00001	0.0827 $\pm$ 0.00002	0.0030$\pm$0.00000	4.08$\times$	6.86$\times$	27.98$\times$
memoryOpStatsPass	F	3/9	67%	0.0118 $\pm$ 0.00001	0.0795 $\pm$ 0.00004	0.0038$\pm$0.00000	3.10$\times$	6.73$\times$	20.87$\times$
throughputModelPass	F	3/9	67%	0.0121 $\pm$ 0.00001	0.0789 $\pm$ 0.00002	0.0030$\pm$0.00000	4.10$\times$	6.52$\times$	26.68$\times$
verifyIR	F	9/9	0%	0.0123 $\pm$ 0.00041	0.0705 $\pm$ 0.00006	0.0034$\pm$0.00000	3.60$\times$	5.71$\times$	20.59$\times$
stripSideEffectsPass	M	7/9	22%	0.0795$\pm$0.00105	0.1860 $\pm$ 0.00101	0.0900 $\pm$ 0.00013	0.88 $\times$	2.34$\times$	2.07$\times$
targetReturnPass	M	9/9	0%	0.0787$\pm$0.00120	0.1856 $\pm$ 0.00028	0.0897 $\pm$ 0.00123	0.88 $\times$	2.36$\times$	2.07$\times$
Total				0.2845	1.091	0.2191	1.30$\times$	3.83$\times$	4.98$\times$
Geomean						3.21$\times$	4.97$\times$	15.95$\times$	

Table 6: Per-pass performance for `DBQuery`, ADT has 15 fields, SoA uses 13 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across `-iterate` runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Am/Sm	Ai/Am	Ai/Sm
countJoins	F	3/15	80%	0.0186 $\pm$ 0.00001	0.0283 $\pm$ 0.00111	0.0109$\pm$0.00001	1.71$\times$	1.52$\times$	2.60$\times$
filterSelectivitySkew	F	5/15	67%	0.0184 $\pm$ 0.00001	0.0278 $\pm$ 0.00001	0.0124$\pm$0.00068	1.49$\times$	1.51$\times$	2.24$\times$
hashJoinPressure	F	5/15	67%	0.0201 $\pm$ 0.00073	0.0322 $\pm$ 0.00001	0.0129$\pm$0.00001	1.55$\times$	1.60$\times$	2.49$\times$
sumCost	F	6/15	60%	0.0185 $\pm$ 0.00079	0.0280 $\pm$ 0.00099	0.0147$\pm$0.00001	1.26$\times$	1.52$\times$	1.90$\times$
sumMemory	F	6/15	60%	0.0186 $\pm$ 0.00001	0.0277 $\pm$ 0.00001	0.0145$\pm$0.00002	1.28$\times$	1.49$\times$	1.90$\times$
sumRows	F	6/15	60%	0.0246 $\pm$ 0.00004	0.0297 $\pm$ 0.00002	0.0216$\pm$0.00003	1.14$\times$	1.21$\times$	1.38$\times$
clearQueryFlags	M	14/15	7%	0.0692$\pm$0.00011	0.0757 $\pm$ 0.00010	0.0820 $\pm$ 0.00007	0.84 $\times$	1.09 $\times$	0.92 $\times$
scaleCosts	M	15/15	0%	0.0684$\pm$0.00011	0.0751 $\pm$ 0.00110	0.0836 $\pm$ 0.00008	0.82 $\times$	1.10 $\times$	0.90 $\times$
Total				0.2564	0.3244	0.2526	1.02 $\times$	1.27$\times$	1.28$\times$
Geomean						1.22$\times$	1.36$\times$	1.67$\times$	

Table 7: Per-pass performance for `DecisionTree`, ADT has 7 fields, SoA uses 6 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across `-iterate` runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Am/Sm	Ai/Am	Ai/Sm
classify Batch	F	5/7	29%	3.170 $\pm$ 0.0056	4.160 $\pm$ 0.0061	2.604$\pm$0.0109	1.22$\times$	1.31$\times$	1.60$\times$
classify Depth	F	4/7	43%	0.0454 $\pm$ 0.00017	0.0593 $\pm$ 0.00001	0.0369$\pm$0.00007	1.23$\times$	1.31$\times$	1.61$\times$
classify tree	F	5/7	29%	0.0453 $\pm$ 0.00098	0.0593 $\pm$ 0.00107	0.0366$\pm$0.00007	1.24$\times$	1.31$\times$	1.62$\times$
countFeatureUses	F	3/7	57%	0.0484 $\pm$ 0.00091	0.1407 $\pm$ 0.00107	0.0322$\pm$0.00002	1.50$\times$	2.91$\times$	4.36$\times$
countLeaves	F	2/7	71%	0.0456 $\pm$ 0.00001	0.1395 $\pm$ 0.00007	0.0248$\pm$0.00002	1.84$\times$	3.06$\times$	5.63$\times$
countNodes	F	2/7	71%	0.0460 $\pm$ 0.00124	0.1391 $\pm$ 0.00116	0.0264$\pm$0.00002	1.74$\times$	3.03$\times$	5.28$\times$
countSmallLeaves	F	3/7	57%	0.0496 $\pm$ 0.00003	0.1347 $\pm$ 0.00132	0.0326$\pm$0.00004	1.52$\times$	2.71$\times$	4.13$\times$
inferenceCost	F	2/7	71%	0.0491 $\pm$ 0.00095	0.1377 $\pm$ 0.00020	0.0291$\pm$0.00093	1.69$\times$	2.80$\times$	4.73$\times$
sumImpurity	F	3/7	57%	0.0484 $\pm$ 0.00001	0.1379 $\pm$ 0.00196	0.0324$\pm$0.00002	1.49$\times$	2.85$\times$	4.26$\times$
sumPathLengths	F	3/7	57%	0.0507 $\pm$ 0.00107	0.1372 $\pm$ 0.00132	0.0338$\pm$0.00002	1.50$\times$	2.70$\times$	4.06$\times$
sumSamples	F	3/7	57%	0.0465 $\pm$ 0.00002	0.1335 $\pm$ 0.00006	0.0324$\pm$0.00002	1.44$\times$	2.87$\times$	4.12$\times$
treeDepth	F	2/7	71%	0.0495 $\pm$ 0.00002	0.1378 $\pm$ 0.00215	0.0281$\pm$0.00001	1.76$\times$	2.78$\times$	4.91$\times$
Total				3.695	5.517	2.950	1.25$\times$	1.49$\times$	1.87$\times$
Geomean						1.50$\times$	2.35$\times$	3.52$\times$	

Table 8: Per-pass performance for `DomTree`, ADT has 15 fields, SoA uses 14 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across `-iterate` runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Am/Sm	Ai/Am	Ai/Sm
count styled	F	3/15	80%	0.0341 $\pm$ 0.00001	0.0687 $\pm$ 0.00108	0.0114$\pm$0.00002	3.00$\times$	2.01$\times$	6.04$\times$
find max Bottom	F	5/15	67%	0.0341 $\pm$ 0.00001	0.0682 $\pm$ 0.00002	0.0228$\pm$0.00003	1.50$\times$	2.00$\times$	2.99$\times$
SumArea	F	6/15	60%	0.0354 $\pm$ 0.00002	0.0690 $\pm$ 0.00003	0.0183$\pm$0.00004	1.94$\times$	1.95$\times$	3.78$\times$
sumTextWidth	F	3/15	80%	0.0342 $\pm$ 0.00002	0.0692 $\pm$ 0.00117	0.0116$\pm$0.00001	2.94$\times$	2.03$\times$	5.96$\times$
computeWidths	M	14/15	7%	0.0293$\pm$0.00127	0.0301 $\pm$ 0.00004	0.0441 $\pm$ 0.00004	0.66 $\times$	1.03 $\times$	0.68 $\times$
scaleLayout	M	15/15	0%	0.0285$\pm$0.00004	0.0299 $\pm$ 0.00004	0.0344 $\pm$ 0.00004	0.83 $\times$	1.05 $\times$	0.87 $\times$
Total				0.1956	0.3351	0.1426	1.37$\times$	1.71$\times$	2.35$\times$
Geomean						1.55$\times$	1.61$\times$	2.49$\times$	

Table 9: Per-pass performance for **KDTree**, ADT has 17 fields, SoA uses 16 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Am/Sm	Ai/Am	Ai/Sm
countInRange tight_box	F	13/17	24%	0.0282 $\pm$ 0.00001	0.0403 $\pm$ 0.00002	0.0198$\pm$0.00082	1.42$\times$	1.43$\times$	2.03$\times$
Find nearest Neighbour	F	13/17	24%	0.0278 $\pm$ 0.00001	0.0394 $\pm$ 0.00001	0.0212$\pm$0.00001	1.31$\times$	1.41$\times$	1.86$\times$
photonMappingBenchmark	F	12/17	29%	0.0422 $\pm$ 0.00001	0.0482 $\pm$ 0.00112	0.0408$\pm$0.00002	1.03 $\times$	1.14$\times$	1.18$\times$
pointCloudNeighborhood	F	11/17	35%	0.0250 $\pm$ 0.00001	0.0386 $\pm$ 0.00002	0.0165$\pm$0.00001	1.51$\times$	1.54$\times$	2.34$\times$
sumMassInRange	F	14/17	18%	0.0276 $\pm$ 0.00064	0.0404 $\pm$ 0.00014	0.0200$\pm$0.00001	1.38$\times$	1.47$\times$	2.02$\times$
twoPointCorrelation bin_8_16	F	11/17	35%	0.0289 $\pm$ 0.00001	0.0392 $\pm$ 0.00001	0.0209$\pm$0.00001	1.38$\times$	1.36$\times$	1.87$\times$
Total				0.1796	0.2460	0.1394	1.29$\times$	1.37$\times$	1.77$\times$
Geomean						1.33$\times$	1.39$\times$	1.84$\times$	

Table 10: Per-pass performance for **LinearListReduction**, ADT has 11 fields, SoA uses 11 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Am/Sm	Ai/Am	Ai/Sm
reduction	F	2/11	82%	0.0297 $\pm$ 0.00001	0.1349 $\pm$ 0.00092	0.0050$\pm$0.00000	5.95$\times$	4.55$\times$	27.10$\times$
Total				0.0297	0.1349	0.0050	5.95$\times$	4.55$\times$	27.10$\times$
Geomean						5.95$\times$	4.55$\times$	27.10$\times$	

Table 11: Per-pass performance for **List**, ADT has 2 fields, SoA uses 2 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Am/Sm	Ai/Am	Ai/Sm
length List	F	1/2	50%	0.0465 $\pm$ 0.00101	<i>OOM</i>	0.0263$\pm$0.00018	1.77$\times$	–	–
sumList	F	2/2	0%	0.1378$\pm$0.03358	<i>OOM</i>	0.1400 $\pm$ 0.00127	0.98 $\times$	–	–
sumList tail recursive	F	2/2	0%	0.1388 $\pm$ 0.00005	<i>OOM</i>	0.1358$\pm$0.00028	1.02 $\times$	–	–
add1 List	M	2/2	0%	0.2498$\pm$0.01204	<i>OOM</i>	0.3387 $\pm$ 0.00944	0.74 $\times$	–	–
Total				0.5729	–	0.6407	0.89 $\times$	–	–
Geomean						1.07 $\times$	–	–	

Table 12: Per-pass performance for **MonoTree**, ADT has 3 fields, SoA uses 2 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Am/Sm	Ai/Am	Ai/Sm
sumTree	F	3/3	0%	0.0091$\pm$0.00001	0.0401 $\pm$ 0.00189	0.0115 $\pm$ 0.00001	0.79 $\times$	4.40$\times$	3.49$\times$
sumTree TailRec	F	3/3	0%	0.0085$\pm$0.00000	0.0403 $\pm$ 0.00001	0.0112 $\pm$ 0.00001	0.76 $\times$	4.72$\times$	3.60$\times$
add1Tree	M	3/3	0%	0.0459 $\pm$ 0.00109	0.0603 $\pm$ 0.00157	0.0425$\pm$0.00004	1.08 $\times$	1.31$\times$	1.42$\times$
Total				0.0635	0.1407	0.0652	0.97 $\times$	2.21$\times$	2.16$\times$
Geomean						0.87 $\times$	3.01$\times$	2.61$\times$	

Table 13: Per-pass performance for **ObjectGraph**, ADT has 5 fields, SoA uses 4 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Am/Sm	Ai/Am	Ai/Sm
countLargeItems	F	3/5	40%	0.0135 $\pm$ 0.00054	0.0444 $\pm$ 0.00002	0.0106$\pm$0.00070	1.27$\times$	3.29$\times$	4.19$\times$
countMarked	F	3/5	40%	0.0138 $\pm$ 0.00001	0.0444 $\pm$ 0.00001	0.0106$\pm$0.00002	1.31$\times$	3.22$\times$	4.20$\times$
countSurvivors	F	4/5	20%	0.0145 $\pm$ 0.00001	0.0447 $\pm$ 0.00002	0.0131$\pm$0.00000	1.10$\times$	3.09$\times$	3.41$\times$
deadBytes	F	4/5	20%	0.0144 $\pm$ 0.00001	0.0437 $\pm$ 0.00018	0.0126$\pm$0.00000	1.15$\times$	3.03$\times$	3.47$\times$
liveBytes	F	4/5	20%	0.0142 $\pm$ 0.00001	0.0432 $\pm$ 0.00001	0.0126$\pm$0.00001	1.13$\times$	3.04$\times$	3.43$\times$
sumObjIds	F	3/5	40%	0.0131 $\pm$ 0.00001	0.0431 $\pm$ 0.00002	0.0104$\pm$0.00002	1.27$\times$	3.28$\times$	4.16$\times$
totalHeapSize	F	3/5	40%	0.0137 $\pm$ 0.00001	0.0430 $\pm$ 0.00001	0.0108$\pm$0.00002	1.27$\times$	3.13$\times$	3.97$\times$
sweepUnmarked	M	5/5	0%	0.0709$\pm$0.00009	0.0881 $\pm$ 0.00010	0.0808 $\pm$ 0.00114	0.88 $\times$	1.24$\times$	1.09 $\times$
touchHotObjects	M	5/5	0%	0.0712$\pm$0.00005	0.0870 $\pm$ 0.00104	0.0805 $\pm$ 0.00003	0.88 $\times$	1.22$\times$	1.08 $\times$
Total				0.2394	0.4814	0.2419	0.99 $\times$	2.01$\times$	1.99$\times$
Geomean						1.13$\times$	2.56$\times$	2.89$\times$	

Table 14: Per-pass performance for **PiecewiseFunctions**, ADT has 8 fields, SoA uses 7 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Am/Sm	Ai/Am	Ai/Sm
autorefineMaxLevel	F	4/8	50%	0.0230 $\pm$ 0.00001	0.0462 $\pm$ 0.00001	0.0195$\pm$0.00002	1.18$\times$	2.01$\times$	2.37$\times$
compressMass	F	3/8	62%	0.0210 $\pm$ 0.00053	0.0467 $\pm$ 0.00002	0.0115$\pm$0.00001	1.83$\times$	2.22$\times$	4.05$\times$
lbDeuxLoadProxy	F	5/8	38%	0.0229 $\pm$ 0.00001	0.0485 $\pm$ 0.00110	0.0178$\pm$0.00002	1.29$\times$	2.12$\times$	2.73$\times$
norm2Estimate	F	5/8	38%	0.0264 $\pm$ 0.00003	0.0559 $\pm$ 0.00002	0.0176$\pm$0.00001	1.50$\times$	2.12$\times$	3.18$\times$
pmapCutHistogram	F	4/8	50%	0.0338 $\pm$ 0.00003	0.0570 $\pm$ 0.00002	0.0124$\pm$0.00001	2.72$\times$	1.69$\times$	4.59$\times$
truncateTolViolations	F	3/8	62%	0.0216 $\pm$ 0.00003	0.0478 $\pm$ 0.00081	0.0113$\pm$0.00001	1.91$\times$	2.21$\times$	4.22$\times$
addConstPW	M	8/8	0%	0.1172$\pm$0.00163	0.1296 $\pm$ 0.00210	0.1281 $\pm$ 0.00246	0.92 $\times$	1.11$\times$	1.01 $\times$
diffPW	M	8/8	0%	0.1197$\pm$0.00121	0.1322 $\pm$ 0.00108	0.1291 $\pm$ 0.00111	0.93 $\times$	1.10$\times$	1.02 $\times$
Total				0.3855	0.5640	0.3472	1.11$\times$	1.46$\times$	1.62$\times$
Geomean						1.44$\times$	1.76$\times$	2.53$\times$	

Table 15: Per-pass performance for **TernaryTree**, ADT has 5 fields, SoA uses 3 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Am/Sm	Ai/Am	Ai/Sm
sum tree	F	5/5	0%	0.0299 $\pm$ 0.00006	0.0506 $\pm$ 0.00001	0.0288$\pm$0.00102	1.04 $\times$	1.70$\times$	1.76$\times$
add 1 tree	M	5/5	0%	0.0835 $\pm$ 0.00009	0.0930 $\pm$ 0.00106	0.0774$\pm$0.00125	1.08 $\times$	1.11$\times$	1.20$\times$
Total				0.1134	0.1436	0.1062	1.07 $\times$	1.27$\times$	1.35$\times$
Geomean						1.06 $\times$	1.37$\times$	1.45$\times$	

Table 16: Per-pass performance for **Trie**, ADT has 10 fields, SoA uses 9 buffers (mutable + immutable cursors). Times are median per iteration (s); $\pm$ shows standard error of the mean across `-iterate` runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	Am	Ai	Sm	Am/Sm	Ai/Am	Ai/Sm
autocompleteTopKProxy	F	5/10	50%	0.0152±0.00001	0.0281±0.00001	0.0081±0.00003	1.87 ×	1.85 ×	3.47 ×
countLazyNodes	F	4/10	60%	0.0136±0.00001	0.0286±0.00104	0.0067±0.00001	2.04 ×	2.09 ×	4.27 ×
countTerminals	F	3/10	70%	0.0129±0.00001	0.0266±0.00001	0.0058±0.00001	2.21 ×	2.06 ×	4.56 ×
sumPrefixFreq	F	3/10	70%	0.0131±0.00001	0.0270±0.00001	0.0057±0.00001	2.28 ×	2.07 ×	4.71 ×
sumSubtreeHints	F	3/10	70%	0.0131±0.00001	0.0270±0.00001	0.0057±0.00001	2.29 ×	2.07 ×	4.73 ×
decayTrieStats	M	10/10	0%	0.0743±0.00120	0.0795±0.00082	0.0852±0.00104	0.87×	1.07×	0.93×
resetTraversalState	M	10/10	0%	0.0750±0.00087	0.0806±0.00011	0.0846±0.00010	0.89×	1.07×	0.95×
Total				0.2172	0.2974	0.2018	1.08×	1.37 ×	1.47 ×
Geomean						1.66 ×	1.69 ×	2.80 ×	